H.N.G. University, Patan

M.Sc.(C.A. & I.T.) SEMESTER - IX

902: Mobile Application

1. C reate an android program that will display android life cycle.

1. Definition
2. Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView android:id="@+id/textview"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Activity Lifecycle Demo"
        android:textSize="18sp"
        android:padding="8dp" />
</LinearLayout>
```

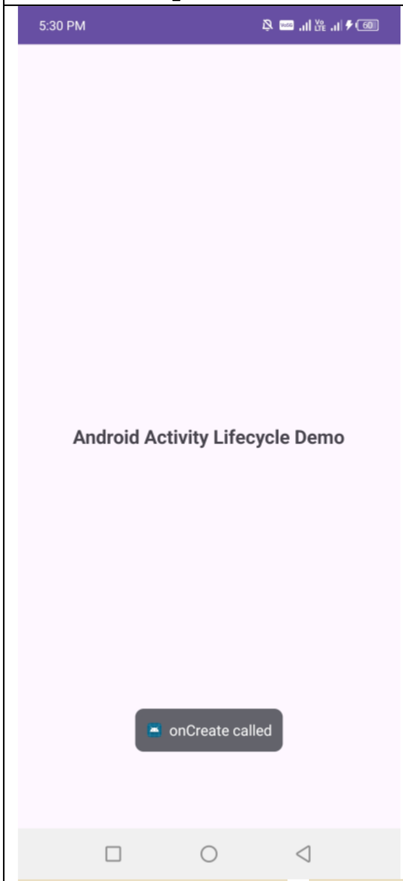
3. Activity.java

```
package com.example.activity_intent;

import android.os.Bundle;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class First extends AppCompatActivity {
    TextView textview;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_first);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
```

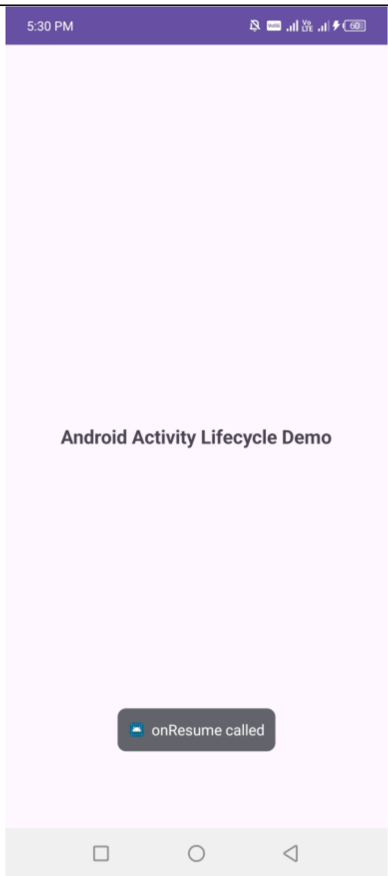
```java
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        textview = findViewById(R.id.textview);
        Toast.makeText(this, "onCreate called",
Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onStart() {
        super.onStart();
        Toast.makeText(this, "onStart called",
Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onPause() {
        super.onPause();
        Toast.makeText(this, "onPause called",
Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onResume() {
        super.onResume();
        Toast.makeText(this, "onResume called",
Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onStop() {
        super.onStop();
        Toast.makeText(this, "onStop called",
Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Toast.makeText(this, "onDestroy called",
Toast.LENGTH_SHORT).show();
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Toast.makeText(this, "onRestart called",
Toast.LENGTH_SHORT).show();
```

```
    }
}
```

4. Output


5:30 PM
Android Activity Lifecycle Demo
onCreate called


5:30 PM
Android Activity Lifecycle Demo
onStart called


5:30 PM
Android Activity Lifecycle Demo
onResume called


5:37 PM
onPause called


5:39 PM
onStop called

2. Create an android program that will display login functionality where username (EmailID) and password should be validated. If username/password will be wrong login button remain disabled otherwise move to next screen (intent) with welcome (username) message.

1. Definition

2. Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".LoginFirst">
    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email"
        android:inputType="textEmailAddress" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Pass"
        android:inputType="textPassword"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:enabled="false"
        android:layout_marginTop="24dp" />
</LinearLayout>
```

3. LoginFirst.java

```
package com.example.myapplication;

import android.content.Intent;
import android.os.Bundle;
import android.text.Editable;
import android.text.TextWatcher;
import android.view.View;
```

```
import android.widget.Button;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class LoginFirst extends AppCompatActivity  implements
TextWatcher{
    EditText etEmail, etPassword;
    Button btnLogin;
    final String correctEmail = "abc";
    final String correctPassword = "123";
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_login_first);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });

        etEmail = findViewById(R.id.etEmail);
        etPassword = findViewById(R.id.etPassword);
        btnLogin = findViewById(R.id.btnLogin);
        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String email = etEmail.getText().toString();
                Intent intent = new
Intent(LoginFirst.this,Second.class);
                intent.putExtra("username", email);
                startActivity(intent);
            }
        });
        etEmail.addTextChangedListener(this);
        etPassword.addTextChangedListener(this);
    }

    @Override
    public void beforeTextChanged(CharSequence s, int start, int
count, int after) {
```

```
    }

    @Override
    public void onTextChanged(CharSequence s, int start, int
before, int count) {
        String emailInput = etEmail.getText().toString().trim();
        String passwordInput =
etPassword.getText().toString().trim();

        boolean valid = emailInput.equals(correctEmail) &&
passwordInput.equals(correctPassword);
        btnLogin.setEnabled(valid);
    }

    @Override
    public void afterTextChanged(Editable s) {

    }
}
```

4. Activity_second.xml

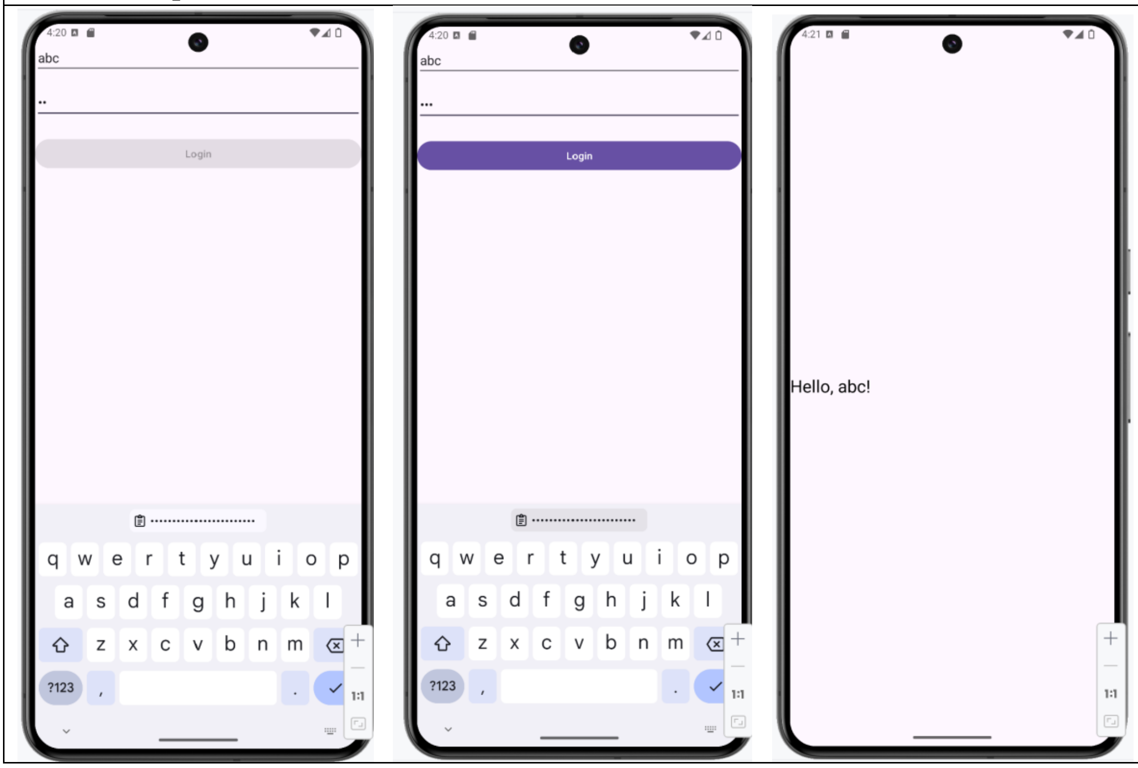
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/textViewName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="24sp"
        android:textColor="#000000" />
</LinearLayout>
```

5. Second.java

```
package com.example.myapplication;

import android.os.Bundle;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class Second extends AppCompatActivity {
    TextView textViewName;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_secoond);
        textViewName = findViewById(R.id.textViewName);

        // Get data from intent
        String name = getIntent().getStringExtra("username");

        textViewName.setText("Hello, " + name + "!");
    }
}
```

6. Output



3. Create an android program that will display use of spinner, where spinner take items and animations from "string.xml" file and selecting an item from spinner-1, image should be changed in image control and selecting an animation from spinner-2 image should be animated.

1. Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:fitsSystemWindows="true"
```

```
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        tools:context=".Spinner">
        <ImageView
            android:id="@+id/imageView"
            android:layout_width="250dp"
            android:layout_height="250dp"
            android:layout_marginTop="32dp"
            android:src="@drawable/p1"
            android:scaleType="fitCenter"/>

        <Spinner
            android:id="@+id/spinnerimage"
            android:layout_width="match_parent"
            android:layout_height="70dp"
            tools:layout_editor_absoluteX="1dp"
            tools:layout_editor_absoluteY="96dp"
            android:entries="@array/strArrayImage"/>
        <Spinner
            android:id="@+id/spinneranim"
            android:layout_width="match_parent"
            android:layout_height="70dp"
            tools:layout_editor_absoluteX="1dp"
            tools:layout_editor_absoluteY="96dp"
            android:entries="@array/strArrayAnim"/>
    </LinearLayout>
```

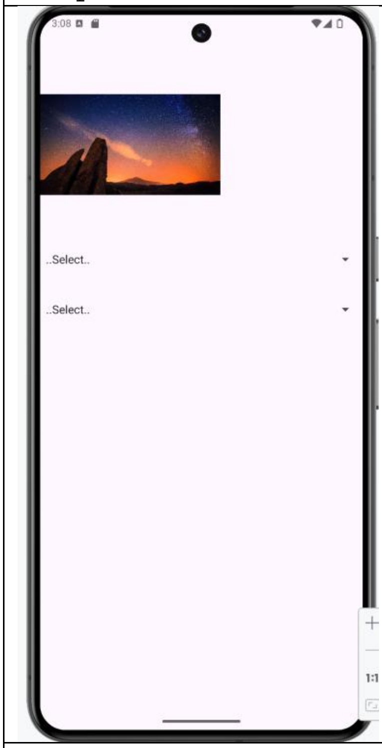
Spinner.java

```
package com.example.myapplication;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.view.animation.Animation;
import android.view.animation.AnimationUtils;
import android.webkit.WebView;
import android.widget.AdapterView;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.LinearLayout;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Spinner extends AppCompatActivity implements
AdapterView.OnItemSelectedListener {
```

```
    LinearLayout l1;
    ImageView iv1;
    android.widget.Spinner spColor,spAnim;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_spinner);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        spColor=findViewById(R.id.spinnerimage);
        spColor.setOnItemSelectedListener(this);
        spAnim=findViewById(R.id.spinneranim);
        spAnim.setOnItemSelectedListener(this);
        l1=findViewById(R.id.main);
        iv1=findViewById(R.id.imageView);
    }
    @Override
    public void onItemSelected(AdapterView<?> parent, View view,
int position, long id) {

       if(parent.getId()==R.id.spinnerimage) {
           String
strColorname=parent.getItemAtPosition(position).toString();
           if (strColorname.equals("p1")) {
               iv1.setImageResource(R.drawable.p1);

           } else if (strColorname.equals("p2")) {
               iv1.setImageResource(R.drawable.p2);

           } else if (strColorname.equals("p3")) {
               iv1.setImageResource(R.drawable.p3);

           } else if (strColorname.equals("p4")) {
               iv1.setImageResource(R.drawable.p4);
           } else
               iv1.setImageResource(R.drawable.p1);
       }
       if(parent.getId()==R.id.spinneranim){
           Toast.makeText(this, ""+position,
Toast.LENGTH_SHORT).show();
           Animation anim = null;
           switch (position) {
```

```
                case 1:
                    anim =
AnimationUtils.loadAnimation(Spinner.this, R.anim.fade);
                    break;
                case 2:
                    anim =
AnimationUtils.loadAnimation(Spinner.this, R.anim.rotate);
                    break;
                case 3:
                    anim =
AnimationUtils.loadAnimation(Spinner.this, R.anim.scale);
                    break;
            }
            if (anim != null) {
                iv1.startAnimation(anim);
            }
        }
    }

    @Override
    public void onNothingSelected(AdapterView<?> parent) {

    }


}
```

Output



String.xml

```xml
<resources>
    <string name="app_name">My Application</string>
    <string name="btnText">Submit</string>
    <string name="strred">RED</string>
    <string name="strBlue">BLUE</string>
    <string-array name="strArrayImage">
        <item>..Select..</item>
        <item>p1</item>
        <item>p2</item>
        <item>p3</item>
        <item>p4</item>
    </string-array>
    <string-array name="strArrayAnim">
        <item>..Select..</item>
        <item>Alpha</item>
        <item>Rotate</item>
        <item>Scale</item>
    </string-array>
</resources>
```

Fade.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<alpha xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="1000"
    android:fromAlpha="0.0"
    android:toAlpha="1.0" />
```

Rotate.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<rotate xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="1000"
    android:fromDegrees="0"
    android:toDegrees="360"
    android:pivotX="50%"
    android:pivotY="50%">
</rotate>
```

Scale.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<scale xmlns:android="http://schemas.android.com/apk/res/android"
    android:duration="1000"
    android:fromXScale="0.5"
    android:toXScale="1.0"
    android:fromYScale="0.5"
    android:toYScale="1.0"
    android:pivotX="50%"
    android:pivotY="50%">
</scale>
```

4. Create an android program that will demonstrate the use of chronometer.

Definition
Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".cronometer">
    <Chronometer
        android:id="@+id/chronometer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="36sp"
        android:format="Time: %s"
        android:gravity="center_horizontal"
        android:layout_marginBottom="32dp" />

    <Button
        android:id="@+id/btnStart"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start" />

    <Button
        android:id="@+id/btnStop"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Stop"
        android:layout_marginTop="16dp" />

    <Button
        android:id="@+id/btnReset"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Reset"
        android:layout_marginTop="16dp" />
</LinearLayout>
```

Activity.java

```
package com.example.myapplication;

import android.os.Bundle;
import android.os.SystemClock;
```

```
import android.view.View;
import android.widget.Button;
import android.widget.Chronometer;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class cronometer extends AppCompatActivity implements
View.OnClickListener{
    Chronometer chronometer;
    Button btnStart, btnStop, btnReset;
    boolean running = false;
    long pauseOffset = 0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_cronometer);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        chronometer = findViewById(R.id.chronometer);
        btnStart = findViewById(R.id.btnStart);
        btnStop = findViewById(R.id.btnStop);
        btnReset = findViewById(R.id.btnReset);

        chronometer.setBase(SystemClock.elapsedRealtime());

        btnStart.setOnClickListener(this);
        btnStop.setOnClickListener(this);
        btnReset.setOnClickListener(this);
    }

    @Override
    public void onClick(View v) {
        if (v == btnStart) {
            if (!running) {
               // chronometer.setBase(System.currentTimeMillis() -
pauseOffset);
                chronometer.setBase(SystemClock.elapsedRealtime()
- pauseOffset);
                chronometer.start();
```

```
                running = true;
            }
        }
        if (v == btnStop) {
            if (running) {
                chronometer.stop();
                //pauseOffset = System.currentTimeMillis() -
chronometer.getBase();
                pauseOffset = SystemClock.elapsedRealtime() -
chronometer.getBase();
                running = false;
            }
        }
        if (v == btnReset) {
            //chronometer.setBase(SystemClock.elapsedRealtime());
            chronometer.setBase(SystemClock.elapsedRealtime());
            pauseOffset = 0;
            if (running) {
                chronometer.stop();
                running = false;
            }
        }

    }
}
```

Output

5. Create an android program that will display use of menu that will change colour of thebackground screen after selecting menu option.

Manifest File

```
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"

android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.MyApplication"
```

```
            tools:targetApi="31">
            <activity
                android:name=".MenuClick"
                android:exported="true"

    android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
                <intent-filter>
                    <action
    android:name="android.intent.action.MAIN" />
                    <category
    android:name="android.intent.category.LAUNCHER" />
                </intent-filter>

            </activity>
        </application>

    </manifest>
```

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MenuClick">

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hello"
        />
</LinearLayout>
```

Activity.java

```
package com.example.myapplication;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.LinearLayout;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
```

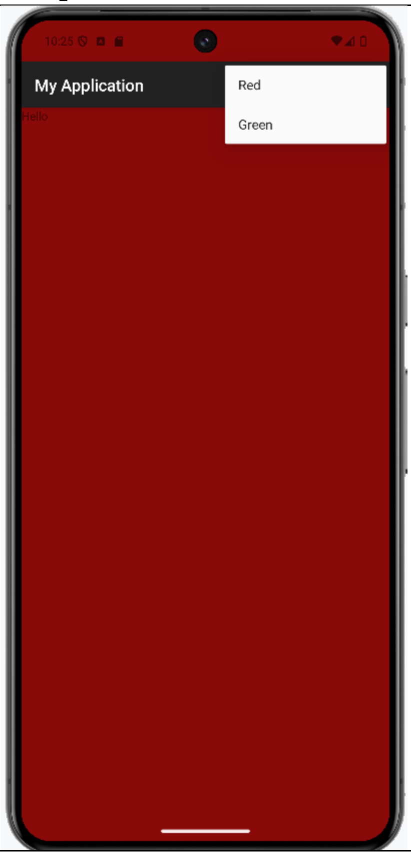
```java
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MenuClick extends AppCompatActivity {
    LinearLayout l1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_menu_click);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        l1=findViewById(R.id.main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.main_menu,menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        if(item.getItemId()==R.id.first){

l1.setBackgroundColor(getResources().getColor(R.color.red));
            Toast.makeText(this, "Red",
Toast.LENGTH_SHORT).show();
        }
        else if(item.getItemId()==R.id.second){

l1.setBackgroundColor(getResources().getColor(R.color.green));
            Toast.makeText(this, "Second",
Toast.LENGTH_SHORT).show();
        }
        return super.onOptionsItemSelected(item);
    }
}
```

Menu.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item android:id="@+id/first"
```

```
        android:orderInCategory="1"
    android:title="Red"/>
     <item android:id="@+id/second"
         android:orderInCategory="2"
         android:title="Green"
         />
</menu>
```

Output



6. Create an android program that will have UI where main screen have list of cars. On selecting of any car, next screen will display no of cars in gallery with car information.

1. String.xml

```
<resources>
    <string-array name="strCar">
        <item>Car 1</item>
        <item>Car 2</item>
        <item>Car 3</item>

    </string-array>

</resources>
```

2. Pra6.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Prac6">

    <ListView
        android:id="@+id/listCar"
        android:layout_width="409dp"
        android:layout_height="729dp"
        tools:layout_editor_absoluteX="1dp"
        tools:layout_editor_absoluteY="1dp" />
</LinearLayout>
```

3. Pra6.java

```java
package com.example.activity_intent;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Prac6 extends AppCompatActivity
        implements AdapterView.OnItemClickListener {
    ListView lCar;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_prac6);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
```

```
                return insets;
            });
            lCar=findViewById(R.id.listCar);
            String[] strCar=getResources().getStringArray
                    (R.array.strCar);
            ArrayAdapter<String> adapter=
                    new ArrayAdapter<>(this,
                            android.R.layout.simple_list_item_1,
                            strCar);
            lCar.setAdapter(adapter);
            lCar.setOnItemClickListener(this);
        }
        @Override
        public void onItemClick(AdapterView<?> parent, View view, int
    position, long id) {
            Intent i = new Intent(Prac6.this,Prac6_2.class);
            String
    stringCar=parent.getItemAtPosition(position).toString();
            i.putExtra("carName",stringCar);
            startActivity(i);
            Toast.makeText(this, stringCar,
    Toast.LENGTH_SHORT).show();
        }
    }
```

4. Prac6_2.xml

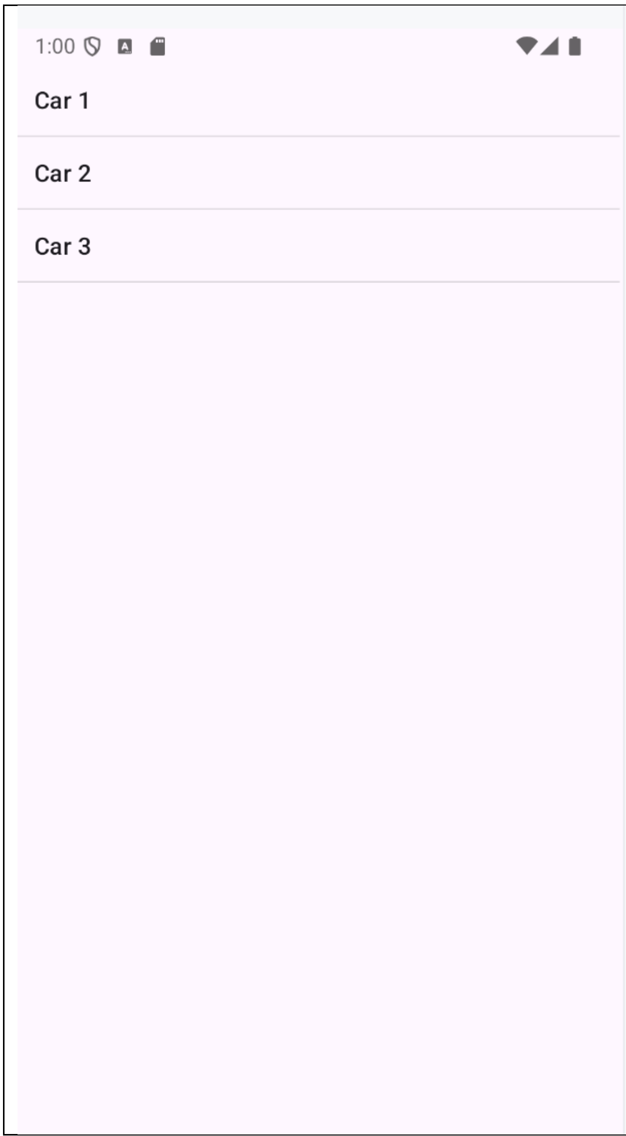
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Prac6_2">
    <Gallery
        android:id="@+id/gallery"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:spacing="10dp" />
    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="TextView"
        tools:layout_editor_absoluteX="37dp"
        tools:layout_editor_absoluteY="48dp" />
</LinearLayout>
```

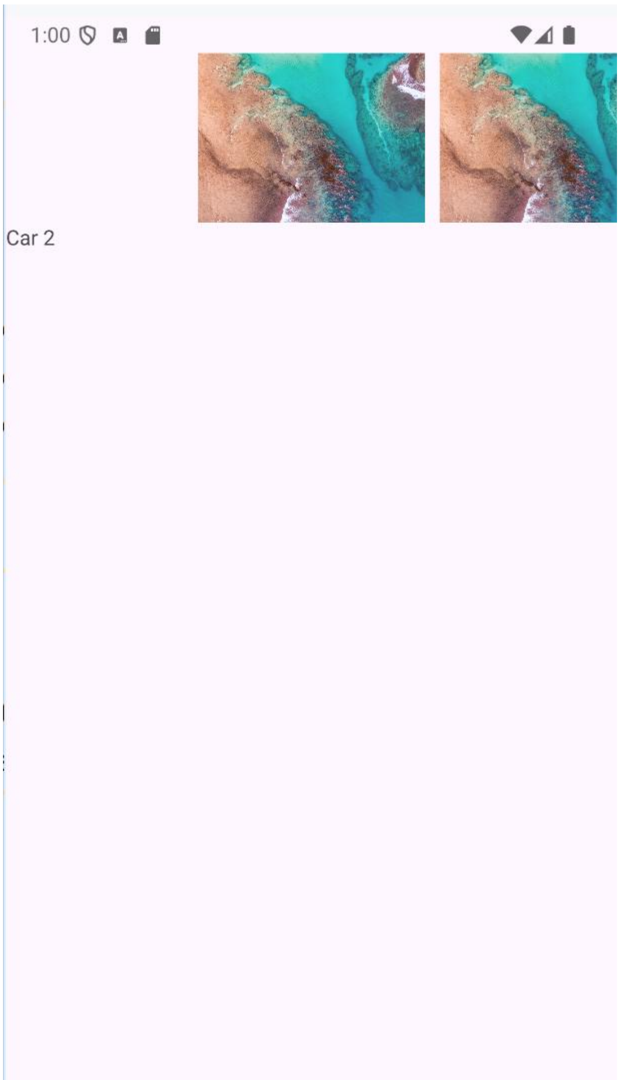
5. Prac6_2.java

```
package com.example.activity_intent;

import android.content.Intent;
import android.os.Bundle;

import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.Gallery;
import android.widget.ImageView;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Prac6_2 extends AppCompatActivity {
    TextView tv1;
    ImageView imageView;
    int[] carImages;
    int[] car1Images = {R.drawable.d1, R.drawable.d2, R.drawable.d3};
    int[] car2Images = {R.drawable.d4, R.drawable.d5, R.drawable.d6};
    int[] car3Images = {R.drawable.d7, R.drawable.d8, R.drawable.d9};
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_prac62);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v,
insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        tv1 =findViewById(R.id.textView);
        Intent i =getIntent();
        String strCarName=i.getStringExtra("carName").toString();
        tv1.setText(strCarName);

        // Assign corresponding image set
        if (strCarName.equals("Car 1")) {
            carImages = car1Images;
        } else if (strCarName.equals("Car 2")) {
            carImages = car2Images;
```

```java
        } else if (strCarName.equals("Car 3")) {
            carImages = car3Images;
        } else {
            // default set
            carImages = new int[]{R.drawable.d1};
        }
        Gallery gallery = findViewById(R.id.gallery);
        gallery.setAdapter(new ImageAdapter());
    }
    public class ImageAdapter extends BaseAdapter {
        @Override
        public int getCount() {
            return carImages.length;
        }
        @Override
        public Object getItem(int position) {
            return carImages[position];
        }
        @Override
        public long getItemId(int position) {
            return position;
        }

        @Override
        public View getView(int position, View convertView, ViewGroup
parent) {
            ImageView imageView = new ImageView(Prac6_2.this);
            imageView.setImageResource(carImages[position]);
            imageView.setLayoutParams(new Gallery.LayoutParams(400,
300));
            imageView.setScaleType(ImageView.ScaleType.CENTER_CROP);
            return imageView;
        }
    }

}
```

Output





7. Create an android program that will read phonebook contacts using content providers and display in list.

1. Manifest.xml

```
<uses-permission android:name="android.permission.READ_CONTACTS"/>
```

2. Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="12dp">

    <TextView
        android:id="@+id/tvHeader"
        android:layout_width="wrap_content"
```

```xml
            android:layout_height="wrap_content"
            android:text="Phone Contacts"
            android:textSize="20sp"
            android:layout_alignParentTop="true"
            android:layout_centerHorizontal="true"
            android:padding="8dp"/>

        <ListView
            android:id="@+id/listContacts"
            android:layout_width="match_parent"
            android:layout_height="match_parent"
            android:layout_below="@id/tvHeader"
            android:dividerHeight="1dp"/>
    </RelativeLayout>
```

3. Activity.java

```java
package com.example.activity_intent;

import android.content.pm.PackageManager;
import android.database.Cursor;
import android.os.Bundle;
import android.provider.ContactsContract;
import android.widget.ArrayAdapter;
import android.widget.ListView;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.Manifest;
import android.widget.Toast;

import java.util.ArrayList;

public class Phone_contact extends AppCompatActivity {
    private static final int REQ_READ_CONTACTS = 101;
    private ListView listContacts;
    private ArrayList<String> contactsList = new ArrayList<>();
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_phone_contact);
        listContacts = findViewById(R.id.listContacts);

        // Check runtime permission
        if (ContextCompat.checkSelfPermission(this,
```

```
Manifest.permission.READ_CONTACTS)
                != PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this,
                    new String[]{Manifest.permission.READ_CONTACTS},
                    REQ_READ_CONTACTS);
        } else {
            loadContacts();
        }
    }
    private void loadContacts() {
        contactsList.clear();

        Cursor cursor = null;
        try {
            cursor =
getContentResolver().query(ContactsContract.CommonDataKinds.Phone.CONTEN
T_URI,
                    new String[]{

ContactsContract.CommonDataKinds.Phone.DISPLAY_NAME,

ContactsContract.CommonDataKinds.Phone.NUMBER
                    },
                    null,
                    null,
                    ContactsContract.CommonDataKinds.Phone.DISPLAY_NAME
+ " ASC"
            );

            if (cursor != null) {
                int nameIdx =
cursor.getColumnIndex(ContactsContract.CommonDataKinds.Phone.DISPLAY_NAM
E);
                int numIdx =
cursor.getColumnIndex(ContactsContract.CommonDataKinds.Phone.NUMBER);

                while (cursor.moveToNext()) {
                    String name = cursor.getString(nameIdx);
                    String number = cursor.getString(numIdx);
                    contactsList.add(name + " - " + number);
                }
            }
        } catch (SecurityException e) {
            Toast.makeText(this, "Permission denied",
Toast.LENGTH_SHORT).show();
        } finally {
            if (cursor != null) cursor.close();
        }

        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
                android.R.layout.simple_list_item_1,
```

```
                contactsList
        );
        listContacts.setAdapter(adapter);
    }

    @Override
    public void onRequestPermissionsResult(int requestCode,String[]
permissions, int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions,
grantResults);
        if (requestCode == REQ_READ_CONTACTS) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                loadContacts();
            } else {
                Toast.makeText(this, "READ_CONTACTS permission denied",
Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

4. Output

8. Create an android program that will play a media file from the memory card.

1. ManifestFile

```
<uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE"/>
```

2. Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```xml
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:id="@+id/tvInfo"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Play Media from SD Card"
            android:textSize="18sp"
            android:padding="10dp"/>

        <Button
            android:id="@+id/btnPlay"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Choose Music"/>

        <Button
            android:id="@+id/btnPause"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Pause"/>
    </LinearLayout>
```

3. Activity.java

```java
package com.example.activity_intent;

import android.content.Intent;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;

public class PlayMediaFromMemoryCard extends AppCompatActivity {
    MediaPlayer mediaPlayer;
    Button btnPick,btnPause;
    boolean isPaused = false;
    private final ActivityResultLauncher<Intent>
openDocumentLauncher =
            registerForActivityResult(new
ActivityResultContracts.StartActivityForResult(),
                    result -> {
```

```
                        if (result.getResultCode() == RESULT_OK
&& result.getData() != null) {
                            Uri fileUri =
result.getData().getData();
                            playMedia(fileUri);
                        } else {
                            Toast.makeText(this, "No file
selected", Toast.LENGTH_SHORT).show();
                        }
                    });
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);

setContentView(R.layout.activity_play_media_from_memory_card);
        btnPick = findViewById(R.id.btnPlay);
        btnPause=findViewById(R.id.btnPause);
        btnPick.setOnClickListener(v -> pickFile());
        btnPause.setOnClickListener(v -> pauseOrResume());
    }

    private void pickFile() {
        Intent intent = new Intent(Intent.ACTION_OPEN_DOCUMENT);
        intent.setType("audio/*");  // only audio files
        intent.addCategory(Intent.CATEGORY_OPENABLE);
        openDocumentLauncher.launch(intent);
    }
    private void playMedia(Uri uri) {
        try {
            if (mediaPlayer != null) {
                mediaPlayer.release();
            }
            mediaPlayer = new MediaPlayer();
            mediaPlayer.setDataSource(this, uri);
            mediaPlayer.prepare();
            mediaPlayer.start();

            Toast.makeText(this, "Playing...",
Toast.LENGTH_SHORT).show();
        } catch (Exception e) {
            Toast.makeText(this, "Error: " + e.getMessage(),
Toast.LENGTH_LONG).show();
            e.printStackTrace();
        }
    }

    private void pauseOrResume() {
        if (mediaPlayer != null) {
            if (mediaPlayer.isPlaying()) {
                mediaPlayer.pause();
                isPaused = true;
```

```java
                btnPause.setText("Play");
                Toast.makeText(this, "Paused",
Toast.LENGTH_SHORT).show();
            } else if (isPaused) {
                mediaPlayer.start();
                btnPause.setText("Pause");
                isPaused = false;
                Toast.makeText(this, "Resumed",
Toast.LENGTH_SHORT).show();
            }
        } else {
            Toast.makeText(this, "No media playing",
Toast.LENGTH_SHORT).show();
        }
    }
    @Override
    protected void onDestroy() {
        super.onDestroy();
        if (mediaPlayer != null) {
            mediaPlayer.release();
            mediaPlayer = null;
        }
    }
}
```

4. Output

9. Create an android application to read file from the sdcard and display that file content to the screen.

1. Activity.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SdCardReading">
    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter text to save" />
    <Button
```

```
            android:id="@+id/btnSave"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Save to SD Card"
            android:layout_marginTop="15dp"/>

        <Button
            android:id="@+id/btnRead"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Read File from SD Card" />


        <TextView
            android:id="@+id/textView"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginTop="20dp"
            android:text="File content will appear here"
            android:textSize="18sp" />
    </LinearLayout>
```

2. Activity.java

```
package com.example.activity_intent;

import android.content.pm.PackageManager;
import android.os.Bundle;
import android.os.Environment;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;


import android.Manifest;
import android.widget.Toast;

import java.io.BufferedReader;
import java.io.File;
import java.io.FileOutputStream;
import java.io.FileReader;
import java.io.IOException;
```

```java
public class SdCardReading extends AppCompatActivity {
    Button btnRead;
    TextView textView;
    EditText editText;
    Button btnSave;
    private static final int PERMISSION_REQUEST_CODE = 1;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_sd_card_reading);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        btnRead = findViewById(R.id.btnRead);
        textView = findViewById(R.id.textView);

        btnRead.setOnClickListener(v -> {
            // Check Permission
                readFileFromSDCard();
        });

        editText = findViewById(R.id.editText);
        btnSave = findViewById(R.id.btnSave);

        btnSave.setOnClickListener(v -> {
                saveFileToSDCard(editText.getText().toString());
        });
    }

    private void readFileFromSDCard() {
        try {
            // Example: File in Downloads/myfile.txt
            File file = new
File(Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_D
OWNLOADS), "myfile.txt");

            if (file.exists()) {
                StringBuilder text = new StringBuilder();
                BufferedReader br = new BufferedReader(new
FileReader(file));
                String line;
                while ((line = br.readLine()) != null) {
                    text.append(line).append("\n");
                }
                br.close();
```

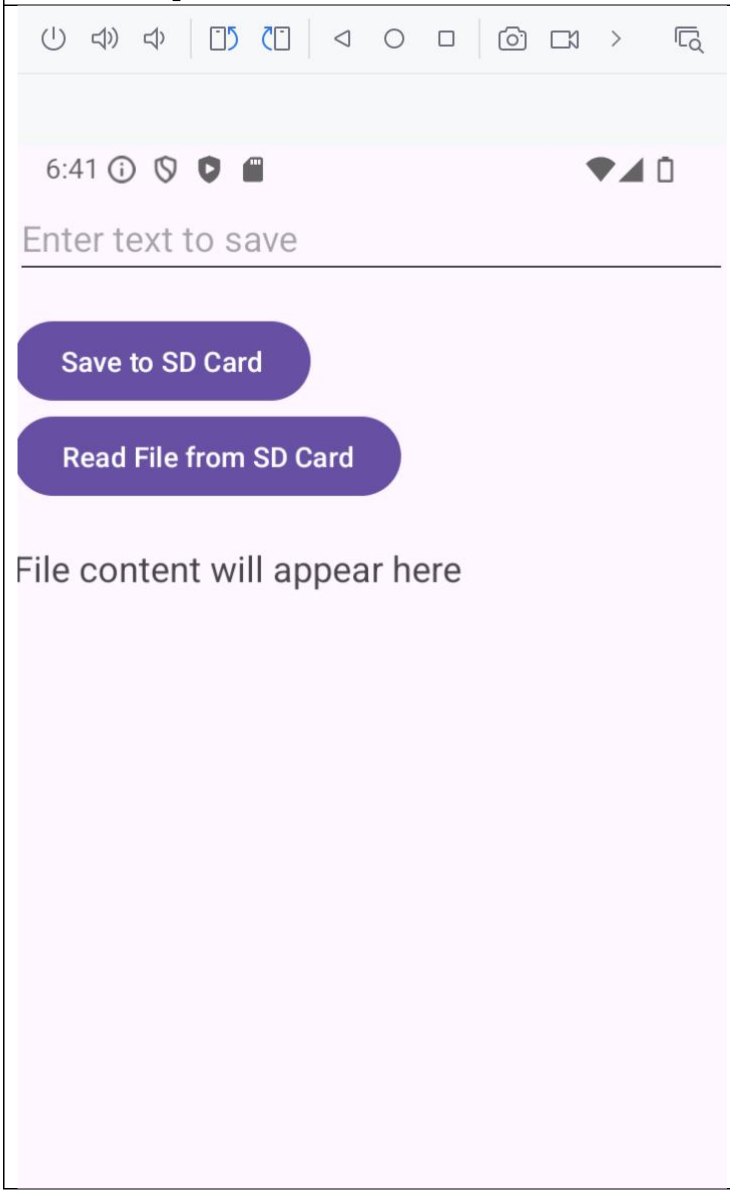
```java
                textView.setText("File Content:\n\n" + text.toString());
            } else {
                textView.setText("File not found: " +
file.getAbsolutePath());
            }
        } catch (Exception e) {
            e.printStackTrace();
            textView.setText("Error reading file: " + e.getMessage());
        }
    }

    private void saveFileToSDCard(String data) {
        try {
            // Save in Downloads/myfile.txt
            File path =
Environment.getExternalStoragePublicDirectory(Environment.DIRECTORY_DOWNLO
ADS);
            File file = new File(path, "myfile.txt");

            FileOutputStream fos = new FileOutputStream(file);
            fos.write(data.getBytes());
            fos.close();

            Toast.makeText(this, "File saved: " + file.getAbsolutePath(),
Toast.LENGTH_LONG).show();
        } catch (IOException e) {
            e.printStackTrace();
            Toast.makeText(this, "Error: " + e.getMessage(),
Toast.LENGTH_SHORT).show();
        }
    }
}
```

3. ManifestFile

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/apk/res/android
"    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE"
/>
 <application android:allowBackup="true"

android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Activity_Intent"
```

```
            tools:targetApi="31">
            <activity
                android:name=".SdCardReading"
                android:exported="true">
                <intent-filter>
                    <action
    android:name="android.intent.action.MAIN" />

                    <category
    android:name="android.intent.category.LAUNCHER" />
                </intent-filter>
            </activity>
        </application>

    </manifest>
```

4. Output



10. Create an android program to perform all database operations like, insert, update, delete, and display.

1. DatabaseHelper.java

```
package com.example.activity_intent;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import androidx.annotation.Nullable;
public class DatabaseHelper extends SQLiteOpenHelper {
    // Database Info
    private static final String DATABASE_NAME = "StudentDB";
    private static final int DATABASE_VERSION = 1;
    // Table Info
    private static final String TABLE_NAME = "students";
    private static final String COL_NO = "student_no";
    private static final String COL_NAME = "name";
    private static final String COL_MOBILE = "mobile";
    private static final String COL_CITY = "city";
    public DatabaseHelper(Context context) {
        super(context,DATABASE_NAME, null,
DATABASE_VERSION);}
    @Override
    public void onCreate(SQLiteDatabase db) {
        String createTable = "CREATE TABLE " + TABLE_NAME +
"("
                + COL_NO + " INTEGER PRIMARY KEY, " +
COL_NAME + " TEXT, "
                + COL_MOBILE + " TEXT, "+ COL_CITY + "
TEXT)";
        db.execSQL(createTable);    }
    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion,
int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }
   public boolean insertStudent(int no, String name, String
mobile, String city) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(COL_NO, no);   values.put(COL_NAME, name);
        values.put(COL_MOBILE, mobile);
values.put(COL_CITY, city);
        long result = db.insert(TABLE_NAME, null, values);
        return result != -1;
```

```
    }

    // Update
    public boolean updateStudent(int no, String name, String mobile, String city) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put(COL_NAME, name);
        values.put(COL_MOBILE, mobile);
        values.put(COL_CITY, city);
        int result = db.update(TABLE_NAME, values, COL_NO + "=?", new String[]{String.valueOf(no)});
        return result > 0;
    }

    // Delete
    public boolean deleteStudent(int no) {
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, COL_NO + "=?", new String[]{String.valueOf(no)});
        return result > 0;
    }

    // Select All
    public Cursor getAllStudents() {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }

    // Search by Student No
    public Cursor searchStudent(int no) {
        SQLiteDatabase db = this.getReadableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME + " WHERE " + COL_NO + "=?",
                new String[]{String.valueOf(no)});
    }


}
```

2. Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
  android:layout_height="match_parent">
    <LinearLayout
        android:orientation="vertical"
        android:padding="16dp"
```

```
            android:layout_width="match_parent"
            android:layout_height="wrap_content">

            <EditText
                android:id="@+id/etNo"
                android:hint="Student No"
                android:inputType="number"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <EditText
                android:id="@+id/etName"
                android:hint="Name"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <EditText
                android:id="@+id/etMobile"
                android:hint="Mobile"
                android:inputType="phone"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <EditText
                android:id="@+id/etCity"
                android:hint="City"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <Button
                android:id="@+id/btnInsert"
                android:text="Insert"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <Button
                android:id="@+id/btnUpdate"
                android:text="Update"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <Button
                android:id="@+id/btnDelete"
                android:text="Delete"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <Button
                android:id="@+id/btnView"
                android:text="View All"
                android:layout_width="match_parent"
```

```
                android:layout_height="wrap_content" />

            <Button
                android:id="@+id/btnSearch"
                android:text="Search by No"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

        </LinearLayout>
    </ScrollView>
```

3. Activity.java

```
package com.example.activity_intent;

import android.content.Intent;
import android.database.Cursor;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Sqllite_operation extends AppCompatActivity {
    DatabaseHelper db;
    EditText etNo, etName, etMobile, etCity;
    Button btnInsert, btnUpdate, btnDelete, btnView, btnSearch;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_sqllite_operation);
        db = new DatabaseHelper(this);
        etNo = findViewById(R.id.etNo);
        etName = findViewById(R.id.etName);
        etMobile = findViewById(R.id.etMobile);
        etCity = findViewById(R.id.etCity);
        btnInsert = findViewById(R.id.btnInsert);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);
        btnView = findViewById(R.id.btnView);
        btnSearch = findViewById(R.id.btnSearch);
        btnInsert.setOnClickListener(v -> {
            if (etNo.getText().toString().isEmpty() ||
etName.getText().toString().isEmpty()
                    || etMobile.getText().toString().isEmpty() ||
etCity.getText().toString().isEmpty()) {
```

```
                Toast.makeText(this, "Enter all fields",
Toast.LENGTH_SHORT).show();
                return;
            }
            boolean inserted = db.insertStudent(
                    Integer.parseInt(etNo.getText().toString()),
                    etName.getText().toString(),
                    etMobile.getText().toString(),
                    etCity.getText().toString());
            Toast.makeText(this, inserted ? "Inserted" : "Failed",
Toast.LENGTH_SHORT).show();
        });

        // Update
        btnUpdate.setOnClickListener(v -> {
            boolean updated = db.updateStudent(
                    Integer.parseInt(etNo.getText().toString()),
                    etName.getText().toString(),
                    etMobile.getText().toString(),
                    etCity.getText().toString());
            Toast.makeText(this, updated ? "Updated" : "Not
Found", Toast.LENGTH_SHORT).show();
        });

        // Delete
        btnDelete.setOnClickListener(v -> {
            boolean deleted =
db.deleteStudent(Integer.parseInt(etNo.getText().toString()));
            Toast.makeText(this, deleted ? "Deleted" : "Not
Found", Toast.LENGTH_SHORT).show();
        });

        // View All
        btnView.setOnClickListener(v -> {
            startActivity(new Intent(Sqllite_operation.this,
ViewAllRecord.class));

        });

        // Search
        btnSearch.setOnClickListener(v -> {
            Cursor cursor =
db.searchStudent(Integer.parseInt(etNo.getText().toString()));
            if (cursor.moveToFirst()) {
                etName.setText(cursor.getString(1));
                etMobile.setText(cursor.getString(2));
                etCity.setText(cursor.getString(3));
                Toast.makeText(this, "Student Found",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Not Found",
```

```
Toast.LENGTH_SHORT).show();
                }
            });
        }
}
```

4. ViewAllRecord.xml

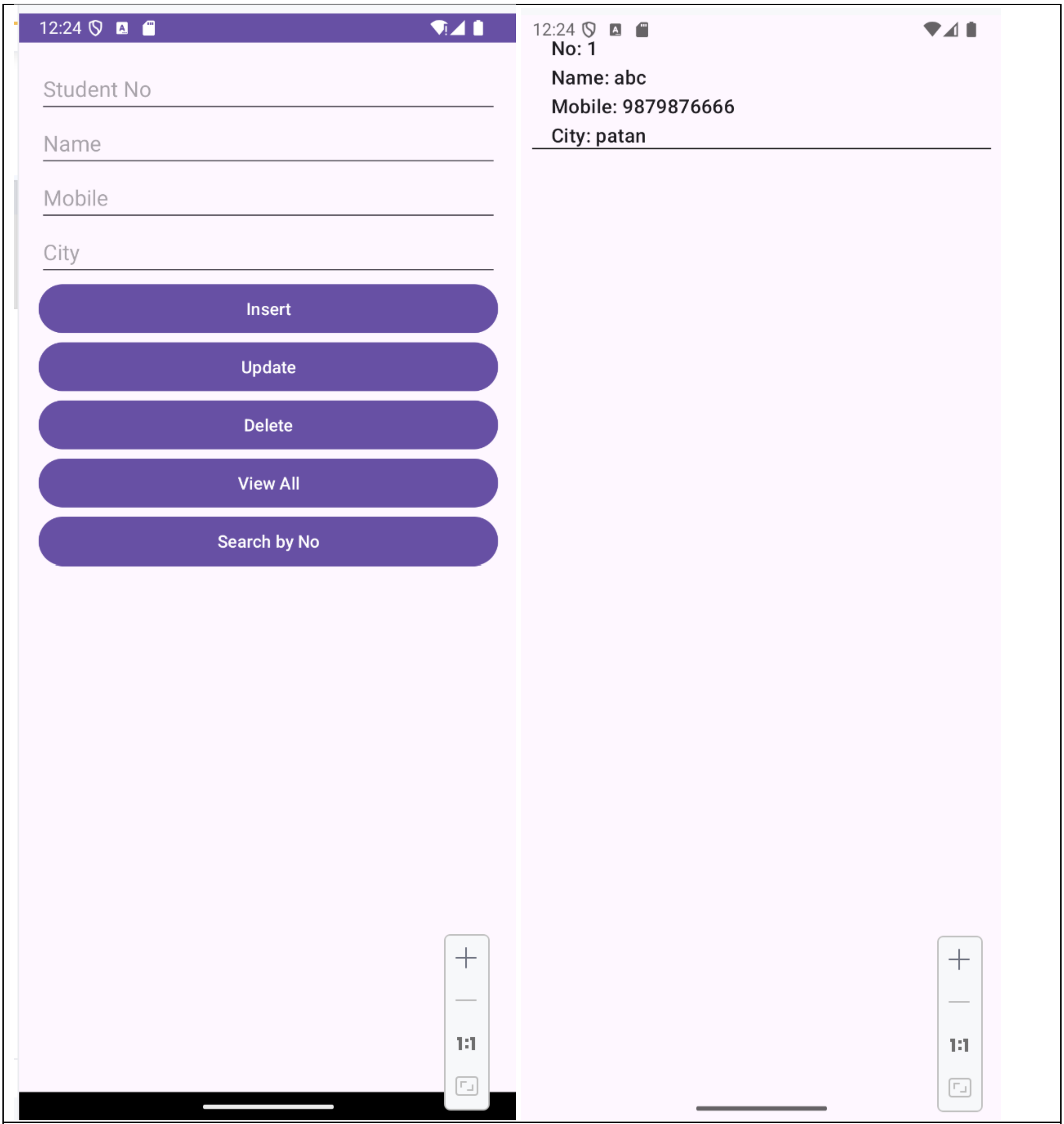
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:divider="#000"
        android:dividerHeight="1dp"/>
</LinearLayout>
```

5. ViewAllRecord.java

```
package com.example.activity_intent;

import android.database.Cursor;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import java.util.ArrayList;

public class ViewAllRecord extends AppCompatActivity {
    DatabaseHelper db;
    ListView listView;
    ArrayList<String> studentList;
    ArrayAdapter<String> adapter;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_view_all_record);

        db = new DatabaseHelper(this);
        listView = findViewById(R.id.listView);
```

```
        studentList = new ArrayList<>();
        adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, studentList);
        listView.setAdapter(adapter);

        loadData();
    }
    private void loadData() {
        Cursor cursor = db.getAllStudents();
        studentList.clear();

        if (cursor.getCount() == 0) {
            Toast.makeText(this, "No Data Found",
Toast.LENGTH_SHORT).show();
            adapter.notifyDataSetChanged();
            return;
        }

        while (cursor.moveToNext()) {
            String student = "No: " + cursor.getInt(0) +
                    "\nName: " + cursor.getString(1) +
                    "\nMobile: " + cursor.getString(2) +
                    "\nCity: " + cursor.getString(3);
            studentList.add(student);
        }

        adapter.notifyDataSetChanged();
    }
}
```

6. Output



11.Create an android program to send message between two emulators and read messages from the any one emulator (mobile). (only work for lower api like 28,29)

Step 1: go to tools -> SDK Managaer-> download API 28 or 29

Step 2: Create AVD with this image

1. ManifestFile.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission
android:name="android.permission.READ_EXTERNAL_STORAGE" />
    <!-- Permission to send SMS -->
    <uses-permission
android:name="android.permission.SEND_SMS"/>
    <!-- Permission to receive SMS -->
    <uses-permission
android:name="android.permission.RECEIVE_SMS"/>
    <uses-permission
android:name="android.permission.READ_SMS"/>
    <uses-feature android:name="android.hardware.telephony"
android:required="false" />
    <application
        android:allowBackup="true"

android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Activity_Intent"
        tools:targetApi="31">

        <activity
            android:name=".sendingMSG"
            android:exported="true">
            <intent-filter>
                <action
android:name="android.intent.action.MAIN" />
                <category
android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
            </activity>
    </application>

</manifest>
```

2. Activity.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
```

```
        android:orientation="vertical"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        tools:context=".sendingMSG">
             <EditText
                android:id="@+id/etNumber"
                android:hint="Enter phone number (e.g., 5556)"
                android:inputType="phone"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <EditText
                android:id="@+id/etMessage"
                android:hint="Enter message"
                android:layout_width="match_parent"
                android:layout_height="wrap_content" />

            <Button
                android:id="@+id/btnSend"
                android:text="Send SMS"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content" />

    </LinearLayout>
```

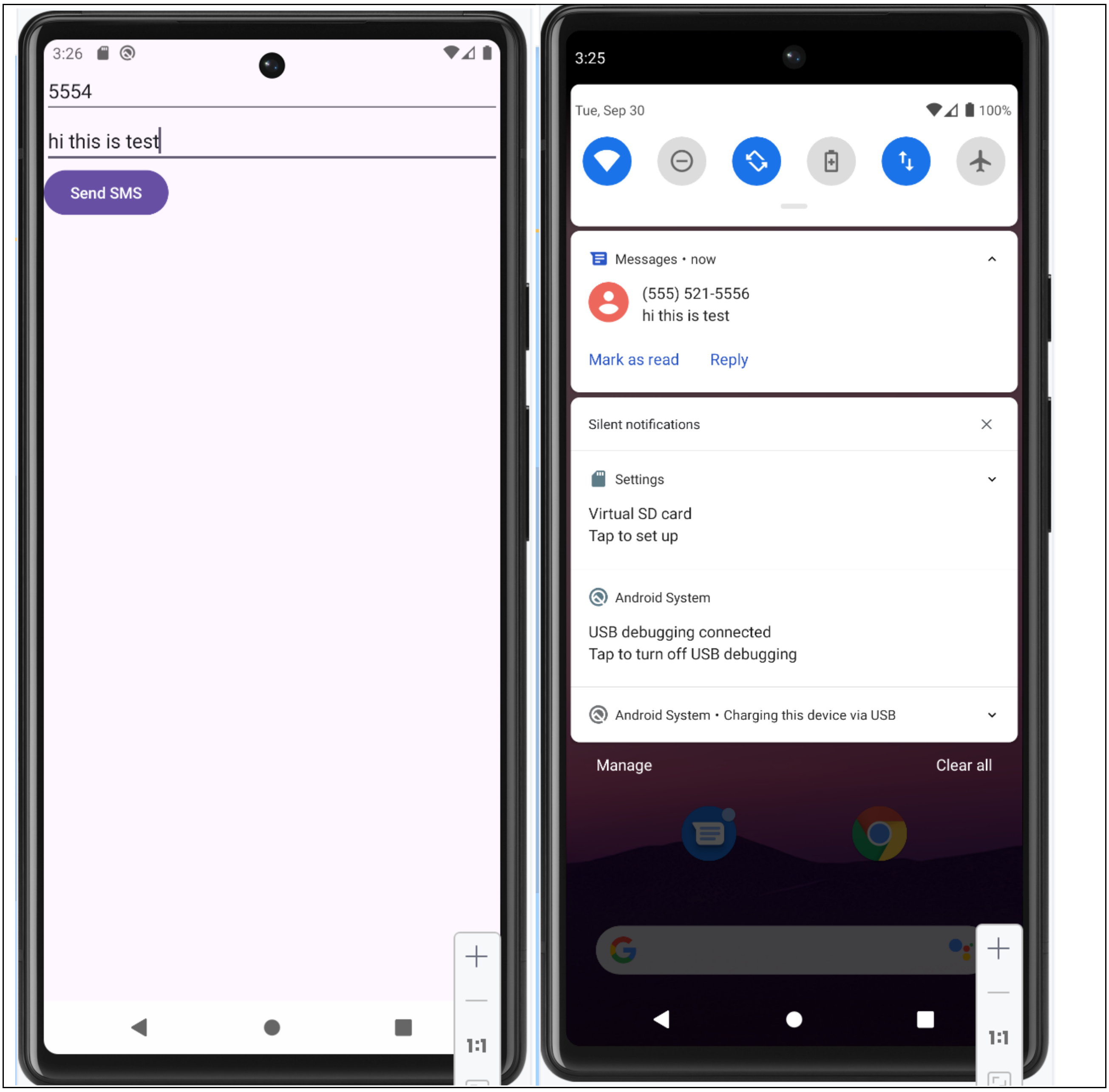
3. Activity.java

```
package com.example.activity_intent;
import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class sendingMSG extends AppCompatActivity {
    EditText etNumber, etMessage;
    Button btnSend;
    private static final int SMS_PERMISSION_CODE = 100;
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_sending_msg);

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });
        etNumber = findViewById(R.id.etNumber);
        etMessage = findViewById(R.id.etMessage);
        btnSend = findViewById(R.id.btnSend);

        btnSend.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if
(ContextCompat.checkSelfPermission(sendingMSG.this,
Manifest.permission.SEND_SMS)
                        != PackageManager.PERMISSION_GRANTED) {

ActivityCompat.requestPermissions(sendingMSG.this,
                            new
String[]{Manifest.permission.SEND_SMS}, 100);
                }else{
                    sendSMS();
                }

            }
        });
    }

    private void sendSMS() {
        String number = etNumber.getText().toString().trim();
        String message = etMessage.getText().toString().trim();

        if (!number.isEmpty() && !message.isEmpty()) {
            try {
                SmsManager smsManager = SmsManager.getDefault();
                smsManager.sendTextMessage(number, null, message,
null, null);
                Toast.makeText(this, "SMS Sent!",
Toast.LENGTH_SHORT).show();
            } catch (Exception e) {
                Toast.makeText(this, "Failed: " + e.getMessage(),
Toast.LENGTH_LONG).show();
            }
```

```
        } else {
            Toast.makeText(this, "Enter number and message",
Toast.LENGTH_SHORT).show();
        }
    }

    @Override
    public void onRequestPermissionsResult(int requestCode,
                                           @NonNull String[]
permissions, @NonNull int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions,
grantResults);

        if (requestCode == 100) {
            if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
                Toast.makeText(this, "SMS Permission Granted",
Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "SMS Permission Denied",
Toast.LENGTH_SHORT).show();
            }
        }
    }
}
```

4. Output



3:26
5554
hi this is test
Send SMS
1:1
3:25
Tue, Sep 30
100%
Messages • now
(555) 521-5556
hi this is test
Mark as read
Reply
Silent notifications
Settings
Virtual SD card
Tap to set up
Android System
USB debugging connected
Tap to turn off USB debugging
Android System • Charging this device via USB
Manage
Clear all
1:1